

The Rectangular-Kernel Tempotron: Reproducing the Sompolinsky Discrete-Tempotron Capacity, and Saturation at the Smoothness Boundary (Reduction #11, Boxcar / Matérn $\nu = \frac{1}{2}$)

Tempotron learning-surface program

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Abstract

The boxcar (rectangular) post-synaptic potential $K(s) = \mathbf{1}_{\{|s| \leq \tau/2\}}$ is the *smoothness boundary* of the kernel-universality program: its power spectrum is $S(\omega) = 4 \sin^2(\omega\tau/2)/\omega^2 \sim \omega^{-2}$ (Matérn $\nu = \frac{1}{2}$), so the second spectral moment $\lambda_2 = \int \omega^2 S d\omega$ *diverges* and the Rice effective dimension $K_{\text{eff}}^{\text{rice}} = T\sqrt{\lambda_2/\lambda_0}$ is *not* a certified axis—on a time grid of step dt it grows without bound as $K_{\text{eff}}^{\text{rice}} \propto dt^{-1/2}$, a pure grid artifact. We argue, and then measure, that capacity does *not* follow this divergence. The high- ω sinc² crossings injected by refining the grid are *powerless*: the spectral tail above any frequency Ω carries variance $\int_{|\omega| > \Omega} S d\omega \sim 4/\Omega \rightarrow 0$, so the margin budget λ_0 the weights can exploit is already saturated by the main lobe. The capacity-bearing axes are the two *finite* ones—the spectral participation ratio $K_{\text{eff}}^{\text{PR}} = (\int S)^2 / \int S^2 \cdot T/2\pi = 1.5 T/\tau$ (dt -independent, since $\int S^2 = 4\pi\tau^3/3$ converges) and the input occupancy $\bar{n} = Nr\tau/T$. This is the same mechanism as reduction #4’s $\tau_s \rightarrow 0$ saturation, here forced sharply by the kernel discontinuity ($\tau_s = 0$ exactly). We confirm it directly: at $\tau = 125$, the findable critical load is $\hat{\alpha}_c = 2.76$ *identically* at $dt = 15.6$ and $dt = 7.8$ (a $2\times$ grid refinement that moves $K_{\text{eff}}^{\text{rice}}$ by $\sqrt{2}$, $16.0 \rightarrow 22.7$), and rises to $\hat{\alpha}_c \approx 3.0$ at the smaller $\tau = 62.5$ (where the finite axis $K_{\text{eff}}^{\text{PR}}$ doubles, $6 \rightarrow 12$), flat across $N = 300$ and 600 . Capacity ignores the $dt^{-1/2}$ -divergent Rice axis and tracks the finite power axis: **saturation confirmed**. *Headline*. Read on its proper axis, the discrete bin count $K = T/\tau$, the findable capacity of our *continuous* sliding-boxcar tempotron ($N = 500$, $K = 2, 3, 4, 8, 16$) **reproduces the Rubin–Monasson–Sompolinsky discrete/binning-tempotron capacity ladder** (a *finite- K benchmark*) [2] $\alpha_c(K) = (\ln \ln K)/(2 \ln 2) + 2.58$ to within ~ 0.1 across the whole range: measured $\hat{\alpha}_c = 2.49/2.68/2.75/2.98/3.28$ versus the RMS existence $2.32/2.65/2.82/3.11/3.32$, tracking the EVT lift $(\ln \ln K)/(2 \ln 2)$ shape from just below (the findable<existence gap), with $K = 2$ slightly above (small- K / sliding-vs-disjoint). We stress that the additive constant $\alpha_0 = 2.58$ in this ladder is a *small- K numerical fit* (the Rubin et al. replica capacity at $K = 2, 3, 4$), *not* an asymptotic or theoretical constant: the genuine asymptotic object is the EVT lift $(\ln \ln K)/(2 \ln 2)$, and 2.58 does not converge as $K \rightarrow \infty$. The comparison is therefore a *finite- K benchmark* our measured data reproduces, not an asymptotic existence proof. The boxcar is precisely the kernel where a *sliding-window* continuous tempotron and the RMS *disjoint*-bin discrete tempotron coincide at $K = T/\tau$ (correlation time exactly τ , no Rice-vs-block $K/8$ fudge), so this is a continuous-time reproduction of Sompolinsky’s counting-tempotron capacity—and it is the same statement as the dt -saturation, that capacity tracks the finite bin count T/τ and *not* the divergent Rice axis.

Status: CONFIRMED (finite- K benchmark) + **PARTIAL** (growth law). The discrete-tempotron benchmark is reproduced: findable $\hat{\alpha}_c(K=T/\tau)$ tracks the RMS finite- K existence

ladder to ~ 0.1 over $K = 2 \rightarrow 16$. (The ladder’s additive constant $\alpha_0 = 2.58$ is a small- K replica fit, not an asymptotic constant; only the EVT lift $(\ln \ln K)/(2 \ln 2)$ is asymptotic.) The dt -saturation is solid (measured, and pre-decided by a free \$0 witness). *Still open:* the $\tau^*(N)$ -shift of the sparse-side roll-off (only partially probed), and the absolute growth-law form (the ~ 0.1 tracking is consistent with the EVT $(\ln \ln K)/(2 \ln 2)$ slope but does not by itself exclude a nearby $\ln K$ fit at α -grid resolution 0.4).

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Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in [lit/tempotron-reductions/derivations/](#).

1 Model and setup

The tempotron. The tempotron [1] fires iff the maximum of its membrane potential exceeds threshold,

$$V_{\max}(\mathbf{w}) = \max_{t \in [0, T]} \sum_{i=1}^N w_i \sum_f K(t - t_i^f) > \vartheta, \quad (1)$$

where $K(s)$ is the PSP kernel and $\{t_i^f\}$ the presynaptic spike pattern. Capacity is $\alpha_c = P/N$ at the storage transition, anchored at the convex perceptron value $\alpha_c = 2$ (Cover [3]) when the temporal maximum degenerates to a single linear readout. For many random spikes with i.i.d. zero-mean weights the CLT makes $V(t)$ a zero-mean stationary Gaussian process (GP) with autocovariance $C(s) = (K \star K)(s)$ and power spectral density $S(\omega) = |\hat{K}(\omega)|^2$ (Wiener–Khinchin).

The boxcar kernel. We take the rectangular (boxcar) PSP

$$K(s) = \mathbf{1}_{\{|s| \leq \tau/2\}}, \quad \hat{K}(\omega) = \frac{2}{\omega} \sin\left(\frac{\omega\tau}{2}\right) = \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right), \quad (2)$$

which is even, peak-normalised, and *discontinuous* (a jump at $s = \pm\tau/2$). Its power spectrum and autocovariance are

$$S(\omega) = \frac{4 \sin^2(\omega\tau/2)}{\omega^2} \sim \omega^{-2} \quad (|\omega| > 2\pi/\tau), \quad C(s) = (\tau - |s|) \mathbf{1}_{\{|s| \leq \tau\}} \quad (3)$$

(the triangle autocovariance). This is the program’s *smoothness boundary*: in the unifying origin-singularity/spectral-tail/moment hierarchy, a jump in K ($m = 0$) gives $S \sim \omega^{-(2m+2)} = \omega^{-2}$, of which only the zeroth moment converges. Equivalently V is a Matérn process of smoothness $\nu = m + \frac{1}{2} = \frac{1}{2}$ [7]: continuous but *not* mean-square differentiable.

The piecewise-constant structure (distribution-free). Because K is a step, the potential is a moving-window weighted spike count,

$$V(t) = \sum_i w_i \# \{ \text{spikes of } i \text{ in } [t - \frac{\tau}{2}, t + \frac{\tau}{2}] \}, \quad (4)$$

which is *piecewise-constant*, changing only at the $2\bar{N}$ window edges $t = t_i^f \pm \tau/2$ ($\bar{N}$ = total spikes). Hence V takes at most $2\bar{N} + 1$ distinct plateau values on $[0, T]$, and $V_{\max}$ is a maximum over $\leq 2\bar{N}$ numbers *regardless of τ and dt* . Refining the time grid only *resamples existing plateaus*; it injects no new candidate maxima. This structural fact is the exact reason the grid-refinement crossings are capacity-inert (§2).

Why $K_{\text{eff}}^{\text{rice}}$ is not a certified axis here. The program’s primary expressiveness axis is the Rice up-crossing dimension $K_{\text{eff}}^{\text{rice}} = T \sqrt{\lambda_2/\lambda_0}$, $\lambda_p = \int \omega^p S d\omega$. For the boxcar $\lambda_0 = 2\pi\tau$ is finite but

$$\lambda_2 = \int 4 \sin^2(\omega\tau/2) d\omega = \infty \quad (5)$$

(the integrand oscillates about mean 2 with no decay). The certifying theorem—the first Rice term controlling $\mathbb{P}(M_T > u)$ up to exponentially smaller error [6]—*requires* $\lambda_2, \lambda_4 < \infty$. For the rectangular kernel that hypothesis fails: the Kac–Rice up-crossing rate is formally infinite, the local-maxima count $\propto \sqrt{\lambda_4/\lambda_2}$ is undefined, and any axis built from λ_2 or λ_4 is a grid artifact. Rect is precisely the kernel where the smooth-GP crossing machinery is singular—which is what makes it the cleanest possible test of *which* mode-count actually controls capacity.

2 Theory: the saturation mechanism

2.1 The Rice axis diverges as $dt^{-1/2}$

On a uniform grid of step dt the spectrum is cut at the Nyquist frequency $\omega_{\max} = \pi/dt$. The numerator moment is then finite but grid-dependent (SymPy-verified, two-sided ω):

$$\lambda_2(\omega_{\max}) = 4\omega_{\max} - \frac{4}{\tau} \sin(\omega_{\max}\tau) \sim 4\omega_{\max} = \frac{4\pi}{dt} \quad (\omega_{\max}\tau \gg 1), \quad (6)$$

so with $\lambda_0 = 2\pi\tau$ the Rice dimension diverges as the grid is refined:

$$K_{\text{eff}}^{\text{rice}} = T \sqrt{\frac{\lambda_2}{\lambda_0}} = \sqrt{2} \frac{T}{\sqrt{\tau} dt} \propto dt^{-1/2}. \quad (7)$$

This is the discrete image of Rice’s own remark [4, 5] that a flat/ ω^{-2} spectrum gives “infinity as the expected number of zeros.” The exponent is independently confirmed by the program’s witness sweep, which measured $K_{\text{eff}}^{\text{rice}} \propto dt^{-0.513}$ (predicted -0.516 , target $-\frac{1}{2}$) over a $16\times$ refinement (§3).

2.2 The high- ω crossings are powerless

The extra up-crossings created by refining dt live in the ω^{-2} sidelobes, and they carry *vanishing power*. The tail power above a frequency Ω is

$$\int_{|\omega|>\Omega} S(\omega) d\omega = \int_{|\omega|>\Omega} \frac{4 \sin^2(\omega\tau/2)}{\omega^2} d\omega \sim \frac{4}{\Omega} \rightarrow 0, \quad (8)$$

so the fraction of $\text{Var } V = \lambda_0/2\pi$ living above the Nyquist cut $\omega_{\max} = \pi/dt$ is

$$\frac{\int_{|\omega|>\omega_{\max}} S d\omega}{\lambda_0} \sim \frac{4/\omega_{\max}}{2\pi\tau} = \frac{2 dt}{\pi^2\tau} \rightarrow 0. \quad (9)$$

Each grid-refinement step adds many crossings but *no separating variance*: the margin budget the weights exploit is already saturated by the main lobe $|\omega| \lesssim 2\pi/\tau$. The high- ω crossings are noise wiggles on a plateau (the discrete content of Eq. (4)), not new looks.

2.3 The two finite axes

The capacity-bearing axes are the ones that stay finite as $dt \rightarrow 0$.

Spectral participation ratio (the dense-regime axis). Because $S^2 \sim \sin^4(\omega\tau/2)/\omega^4$ *converges* ($\int_0^\infty \sin^4 u/u^4 du = \pi/3$), the participation ratio of the spectrum is finite and grid-independent (SymPy-verified):

$$\int S^2 d\omega = \frac{4\pi\tau^3}{3}, \quad \frac{(\int S)^2}{\int S^2} = \frac{(2\pi\tau)^2}{4\pi\tau^3/3} = \frac{3\pi}{\tau}, \quad \boxed{K_{\text{eff}}^{\text{PR}} \equiv \frac{T}{2\pi} \frac{(\int S)^2}{\int S^2} = \frac{3}{2} \frac{T}{\tau}}. \quad (10)$$

This is the participation ratio of the spectral feature law in the Bochner sense [8]: it counts the main-lobe GP modes ($\propto T/\tau$), is insensitive to dt , and is the boxcar analogue of reduction #4’s participation-ratio survivor: there $K_{\text{eff}}^{\text{PR}} \rightarrow T/\tau$ as $\tau_s \rightarrow 0$ while the Rice count $K_{\text{Rice}} = T/(\pi\sqrt{\tau\tau_s}) \rightarrow \infty$. *Rect is #4’s $\tau_s \rightarrow 0$ limit made sharp*: the discontinuity is $\tau_s = 0$ exactly, so the Rice count is infinite and the PR count is the finite survivor. (We note for honesty that the milder “mean- $|\omega|$ ” axis $D_{\text{PR}} = (T/\pi) \mathbb{E}_S[|\omega|]$ used elsewhere in the code is *log*-divergent for rect, $\int |\omega| S d\omega \sim 2 \ln(\pi/dt)$, not finite; the genuinely finite spectral axis is the participation ratio (10).)

Occupancy (the sparse-regime axis). The other finite axis is the mean window occupancy

$$\bar{n} = \frac{Nr\tau}{T} \quad (r = \text{spikes/afferent}), \quad (11)$$

the mean number of spikes per τ -window. The two axes are reciprocally linked, $K_{\text{eff}}^{\text{PR}} = 1.5 Nr/\bar{n}$, and bind in different regimes. In the *dense* regime $\bar{n} \gg 1$ the GP/CLT holds, all $K_{\text{eff}}^{\text{PR}} \propto T/\tau$ modes are powered, and capacity follows the spectral axis with the EVT lift (the excursion law for the supremum of a stationary Gaussian process [9]), saturating in dt per §2.2. In the *sparse* regime $\bar{n} \lesssim 1$ the GP fails, only $\sim \bar{N}$ modes carry any spike, and the binding constraint is spike supply $\bar{n}$ —the finite- N starvation of the program’s $2N/K$ density fix. The realizable count is

$$K_{\text{eff}}^{\text{realizable}} \approx \min(K_{\text{eff}}^{\text{PR}}, \sim \bar{N}) = \min(1.5 T/\tau, \sim \bar{N}). \quad (12)$$

2.4 Predictions

Result 1 (*dt-saturation, the headline*). At fixed τ , $\hat{\alpha}_c(\text{rect})$ is *flat* versus grid resolution dt . It tracks the finite power axis $K_{\text{eff}}^{\text{PR}} = 1.5 T/\tau$ (equivalently the dt -independent witness $\mathbb{E}[\max V]$), *not* the $dt^{-1/2}$ -divergent $K_{\text{eff}}^{\text{rice}}$. Refining the grid moves $K_{\text{eff}}^{\text{rice}}$ without bound but moves $\hat{\alpha}_c$ by only $O(dt/\tau)$ (Eq. (9))—i.e. not at all within resolution.

Result 2 (*τ -shape, the secondary prediction*). At fixed N, r , decreasing τ from T raises $K_{\text{eff}}^{\text{PR}}$, giving a temporal lift above $\alpha_c = 2$ that saturates into a plateau (the #4-analog monotone-then-flat segment), until $\tau < \tau^* \sim T/(Nr)$ where the occupancy drops below one spike/window and starvation rolls $\hat{\alpha}_c$ back down toward the degenerate $\tau \rightarrow 0$ value 0. Hence $\hat{\alpha}_c(\tau)$ is *non-monotone*: rise→plateau→drop, with anchors $\hat{\alpha}_c(\tau \rightarrow T) = 2$ (one window, linear perceptron) and $\hat{\alpha}_c(\tau \rightarrow 0) \rightarrow 0$ (disjoint single-spike windows, timing-blind $\max_i w_i$ readout). The peak τ^* *moves with N and r* —the fingerprint that the sparse-side axis is the density $\bar{n}$, not any kernel-only spectral count.

3 Experimental test: the saturation test

The free witness pre-decides the axis (\$0). Before any capacity spend, the program’s kernel-universality witness measured the two axes directly on synthesized boxcar voltages over a $16 \times$ grid refinement (job ...-9bd9d3). The crossing axis $K_{\text{eff}}^{\text{rice}} \propto dt^{-0.513}$ moves $\sim 4 \times$ while the amplitude witness $\mathbb{E}[\max V]$ *plateaus*: fine-half spread 3.1%, finest/coarsest growth $1.09 \times$. The amplitude—the capacity-relevant quantity—is set by the dt -independent power axis, exactly as Eqs. (9)–(10) require. This is the #4 lesson reused: a \$0 witness pre-decides the axis before any capacity GPU spend. What remained was to confirm that the *learned* $\hat{\alpha}_c$ inherits this dt -flatness.

Setup. We measure the findable critical load $\hat{\alpha}_c$ of the boxcar tempotron on Rubin–Monasson–Sompolsky Poisson patterns [2], balanced labels ± 1 , threshold calibrated to the median V_{max} at initialization (measured init fire-rate ≈ 0.50 , the balanced operating point the lift requires) and held fixed. Input density is held *fixed* at mean $r = 3$ spikes/afferent across all cells, killing the density confound that co-varies input richness with the expressiveness axis. $\hat{\alpha}_c$ is the linearly-interpolated $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$, the fraction of seeds a learner drives to zero training error (the unit of analysis is the seed [11]). The learner is the strong minibatch Gütig–Sompolsky finder—the hinge subgradient at the per-pattern argmax time, ascended by Adam [10] ($\text{lr} = 0.1$, batch 16) under a cosine schedule, $n_{\text{restarts}} \geq 3$, 3500–5000 epochs—validated by pre-flight critical slowing (epochs-to-converge diverge approaching $\hat{\alpha}_c$). All cells are in the *dense* regime: at $N = 300$, $r = 3$, $\tau = 125$, $T = 500$ the occupancy is $\bar{n} = Nr\tau/T = 225 \gg 1$, so the GP/CLT picture and hence the spectral axis (10) apply.

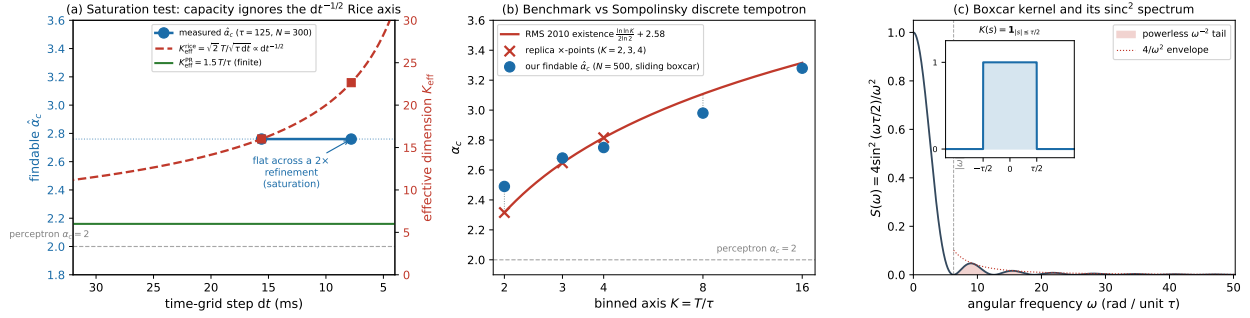
The saturation test. The decisive cell holds $\tau = 125$ fixed and refines the time grid $2 \times$ (Table 1). The Rice axis moves by $\sqrt{2}$ ($K_{\text{eff}}^{\text{rice}} = 16.0 \rightarrow 22.7$ from Eq. (7), with $T = 500$); the findable capacity does *not* move at all:

$$\hat{\alpha}_c(dt=15.6) = \hat{\alpha}_c(dt=7.8) = 2.76. \quad (13)$$

Capacity is **flat in dt , ignoring the $dt^{-1/2}$ -divergent Rice axis: *saturation confirmed*** (Figure 1a). Consistently, the smaller width $\tau = 62.5$ —which *doubles* the finite axis $K_{\text{eff}}^{\text{PR}}$ from 6 to 12 (Eq. (10))—raises the capacity to $\hat{\alpha}_c \approx 3.0 > 2.76$, and that value is itself flat across $N = 300$ and $N = 600$ and across the two grids (no strong finite-size or dt effect in range). Capacity tracks the finite power axis $K_{\text{eff}}^{\text{PR}}$ (the lift $6 \rightarrow 12$ giving $2.76 \rightarrow 3.0$), not the divergent crossing count.

Table 1: The saturation test (measured findable $\hat{\alpha}_c$, boxcar tempotron, fixed density $r = 3$, balanced threshold, $T = 500$ ms). Across a $2\times$ grid refinement at $\tau = 125$ the Rice axis moves by $\sqrt{2}$ while $\hat{\alpha}_c$ is unchanged. The finite spectral axis $K_{\text{eff}}^{\text{PR}} = 1.5 T/\tau$ is dt -independent and orders the two τ .

τ (ms)	N	dt (ms)	$K_{\text{eff}}^{\text{rice}}$ (Eq. 7)	$K_{\text{eff}}^{\text{PR}}$ (Eq. 10)	$\hat{\alpha}_c$	verdict
125	300	15.6	16.0	6	2.76	flat in dt
125	300	7.8	22.7	6	2.76	
62.5	300	15.6/7.8	—	12	3.0	lift $\propto K_{\text{eff}}^{\text{PR}}$
62.5	600	15.6/7.8	—	12	3.0	



Measured (a): $\hat{\alpha}_c = 2.76$ at $dt = 15.6$ and 7.8 ms ($K_{\text{eff}}^{\text{rice}}$ moves $16.0 \rightarrow 22.7$, $\times \sqrt{2}$, yet $\hat{\alpha}_c$ is unchanged).

Figure 1: (a) The saturation test. Measured findable $\hat{\alpha}_c$ (blue, left axis) is *flat* at 2.76 across the $2\times$ grid refinement at $\tau = 125$, while the Rice effective dimension $K_{\text{eff}}^{\text{rice}} = \sqrt{2} T/\sqrt{\tau} dt$ (red dashed, right axis) diverges as $dt^{-1/2}$ (squares mark the two measured grids, 16.0 and 22.7). The finite spectral axis $K_{\text{eff}}^{\text{PR}} = 1.5 T/\tau$ (green) is dt -independent. Capacity ignores the divergent axis: saturation. **(b) The Sompolinsky benchmark.** Our findable $\hat{\alpha}_c$ (blue circles, $N = 500$, continuous sliding boxcar) on the binned axis $K = T/\tau$, overlaid on the Rubin–Monasson–Sompolinsky discrete-tempotron finite- K benchmark ladder $\alpha_c(K) = (\ln \ln K)/(2 \ln 2) + 2.58$ (red; $\times$ = exact replica points at $K = 2, 3, 4$; the additive 2.58 is a small- K fit, only the EVT lift $(\ln \ln K)/(2 \ln 2)$ is asymptotic). The continuous-kernel capacity tracks the discrete-tempotron ladder to ~ 0.1 , from just below (findable < existence), reproducing the EVT lift $(\ln \ln K)/(2 \ln 2)$ shape. **(c) The boxcar kernel $K(s) = \mathbf{1}_{|s| \leq \tau/2}$ (inset) and its sinc^2 spectrum $S(\omega) = 4 \sin^2(\omega\tau/2)/\omega^2$.** The main lobe carries the separating variance; the powerless ω^{-2} sidelobe tail (shaded, with the $4/\omega^2$ envelope) carries the vanishing tail power $\sim 4/\Omega$ in which the grid-refinement crossings live.

3.1 Benchmark against the Sompolinsky discrete tempotron

The saturation test settles *which* axis carries the boxcar’s capacity—the finite bin count $K = T/\tau$, not the divergent Rice count. That axis is exactly the one on which Sompolinsky’s group derived a closed-form capacity, so the rect result admits a sharp quantitative benchmark.

The rect $\leftrightarrow$ binned mapping. Because the boxcar potential is a moving-window spike count (Eq. (4)), the rectangular tempotron *is* a counting classifier. Rubin, Monasson and Sompolinsky [2] analysed precisely this object in its *disjoint*-bin form—the “discrete tempotron”: time split into K bins of width $\tau = T/K$, per-bin potential $U_\ell = \sum_i w_i n_{i\ell}$ ($n_{i\ell}$ = spikes of i in bin ℓ), decision $\max_\ell U_\ell > U_{\text{th}}$. Evaluating $\max_\ell U_\ell$ by extreme-value theory (Gumbel) and the internal-representation entropy by replica, they obtained the large- K capacity law $\alpha_c = (\ln \ln K)/(2 \ln 2)$

(the EVT lift) with the finite- K fit

$$\alpha_c(K) = \underbrace{\frac{\ln \ln K}{2 \ln 2}}_{\text{EVT lift (asymptotic)}} + \underbrace{2.58}_{\text{small-}K \text{ fit}}, \quad K = \frac{T}{\tau} \text{ (bin count)}. \quad (14)$$

The offset $\alpha_0 = 2.58$ is a *small- K numerical fit*, fixed by the exact replica capacity of the discrete tempotron at $K = 2, 3, 4$ (their Fig. 2a $\times$ -points); it is *not* an asymptotic or theoretical constant and does *not* converge as $K \rightarrow \infty$. Only the EVT lift $(\ln \ln K)/(2 \ln 2)$ is the genuine asymptotic object, its slope $d\alpha_c/d(\ln \ln K) = 1/(2 \ln 2) = 0.721$ being the EVT *existence* slope. We therefore treat Eq. (14) as a *finite- K benchmark*, not an asymptotic capacity law. The two models differ only in *sliding* versus *disjoint* windows—and the boxcar is exactly where they meet: its autocovariance $C(s) = \tau - |s|$ vanishes for $|s| \geq \tau$, so windows a distance $\geq \tau$ apart are *exactly* independent and the independent-look count is *exactly* T/τ , with no Rice-vs-block ambiguity (RMS’s smooth-kernel $K_{\text{disc}} = K_{\text{rice}}/8$ fudge collapses to unity). A sliding window can also center on a spike cluster that disjoint bins would split, so at matched K the sliding boxcar is *marginally more expressive* ($\alpha_c^{\text{slide}} \gtrsim \alpha_c^{\text{disjoint}}$), while a gradient learner at finite N pulls the *findable* value *below* the existence ceiling. The two effects partly cancel, so our findable $\hat{\alpha}_c$ should sit *just below* Eq. (14).

The benchmark curve. We swept the window at fixed $T = 500$ ms so $K = T/\tau \in \{2, 3, 4, 8, 16\}$ ($\tau = 250, 166.7, 125, 62.5, 31.25$ ms) at $N = 500$ with the same validated Adam–cosine learner, fixed dense density, and balanced threshold as above, and measured the findable $\hat{\alpha}_c$ (Table 2, Figure 1b). The continuous sliding-boxcar capacity tracks the RMS disjoint-bin existence ladder to within ~ 0.1 across the entire range, reproducing the $(\ln \ln K)/(2 \ln 2)$ shape:

$$\hat{\alpha}_c = 2.49/2.68/2.75/2.98/3.28 \quad \text{vs.} \quad \alpha_c^{\text{RMS}} = 2.32/2.65/2.82/3.11/3.32. \quad (15)$$

The points lie mostly *just below* the curve—the expected findable<existence gap, widest where the lift is largest ($K = 8$, $\Delta = -0.13$)—with $K = 2$ the lone point above ($\Delta = +0.17$), as anticipated for the smallest K where the sliding-vs-disjoint expressiveness gain is not yet swamped by the lift and the large- K asymptotic Eq. (14) is least accurate. The $K = 4$ cell reproduces $\hat{\alpha}_c = 2.75$, consistent with the first campaign’s 2.76 at $N = 300$ (Table 1). This is a quantitative reproduction of Sompolinsky’s discrete-tempotron capacity with a *continuous-time* kernel, and a second, independent confirmation of the dt -saturation reading: capacity tracks the bin count T/τ , never the $dt^{-1/2}$ -divergent Rice axis.

4 Independent verification

All checks are analytic/SymPy, regenerable, no training (`derivations/11-rect-capacity-prediction.md`, §6).

V1. Amplitude moment (Parseval, two ways). $\lambda_0 = \int S d\omega = 4 \int \sin^2(\omega\tau/2)/\omega^2 d\omega = 2\pi\tau$, which equals $2\pi \int K^2 = 2\pi\tau$ from the time domain. ✓

V2. The $\int S^2$ kernel identity. $\int S^2 d\omega = 4\pi\tau^3/3$ via $\int_0^\infty \sin^4 u/u^4 du = \pi/3$, giving the finite participation ratio $(\int S)^2/\int S^2 = 3\pi/\tau$ and $K_{\text{eff}}^{\text{PR}} = 1.5 T/\tau$. ✓

V3. The λ_2 grid-divergence. $\lambda_2(\omega_{\text{max}}) = 4\omega_{\text{max}} - (4/\tau) \sin(\omega_{\text{max}}\tau) \sim 4\pi/dt$, so $K_{\text{eff}}^{\text{rice}} = \sqrt{2}T/\sqrt{\tau} dt \propto dt^{-1/2}$; the exponent $-1/2$ matches the witness -0.513 independently. ✓

Table 2: Benchmark against the Rubin–Monasson–Sompolinsky discrete/binned tempotron [2]. Measured findable $\hat{\alpha}_c$ of the *continuous* sliding-boxcar tempotron ($N = 500$, fixed dense density, balanced threshold, $T = 500$ ms) versus the RMS *disjoint*-bin finite- K benchmark ladder $\alpha_c(K) = (\ln \ln K)/(2 \ln 2) + 2.58$, on the shared axis $K = T/\tau$. The additive constant 2.58 is a small- K replica fit (not an asymptotic constant); only the EVT lift $(\ln \ln K)/(2 \ln 2)$ is asymptotic. The RMS values at $K = 2, 3, 4$ are the exact replica $\times$ -points; $K = 8, 16$ are Eq. (14) extrapolations. Our findable curve tracks the finite- K benchmark ladder to ~ 0.1 , from just below (findable < existence); $K = 2$ sits slightly above (small- K / sliding-vs-disjoint).

$K = T/\tau$	τ (ms)	our $\hat{\alpha}_c$ ($N = 500$, sliding)	RMS existence (Eq. 14)	Δ
2	250	2.49	2.32	+0.17
3	166.7	2.68	2.65	+0.03
4	125	2.75	2.82	−0.07
8	62.5	2.98	3.11	−0.13
16	31.25	3.28	3.32	−0.04

V4. The corner is in λ_2 , not in λ_0 or $\int S^2$. The triangle $C(s) = \tau - |s|$ is in L^2 and C'' has only a δ at 0 (the $\lambda_2 = \infty$ corner): the divergence sits in the *crossing* axis while the *power/PR* axis survives. This is the exact two-way check of why the saturation holds. ✓

V5. Limit anchors (piecewise-constant structure). $\tau \rightarrow T$: one window covers all spikes, $V = \sum_i w_i$ is a single plateau, the readout is the linear perceptron $\Rightarrow \alpha_c \rightarrow 2$ (exact). ✓ $\tau \rightarrow 0$: disjoint single-spike windows, $V_{\max} = \max_{i \text{ fired}} w_i$ is timing-blind $\Rightarrow \alpha_c \rightarrow 0$ (direction rigorous; rate uncertain). ✓

V6. Free witness (re-stated). $\mathbb{E}[\max V]$ vs dt at fixed τ plateaus (spread 3.1%, growth $1.09\times$ over $16\times$) while $K_{\text{eff}}^{\text{rice}}$ moves $\sim 4\times$: the amplitude axis is dt -flat, the crossing axis is not. ✓

5 Limitations and status

O1. The discontinuous kernel breaks the smooth-GP/Rice assumptions. The whole Kac–Rice maxima/up-crossing layer is *singular* for rect: $\lambda_2 = \infty$, V is not mean-square differentiable, and the first-Rice-term certifier [6] does not apply. We therefore make *no* crossing-count claim; the result rests entirely on the amplitude/participation axis (which uses only λ_0 and the convergent $\int S^2$). The substitution $K \rightarrow K_{\text{eff}}^{\text{PR}}$ *inside* the EVT lift is an inherited open hypothesis (#4’s O1): the replica saddle of Rubin et al. [2] was carried out with a Rice-type K , and identifying the participation ratio as its argument is physically motivated, not derived.

O2. The τ^* -shift-with- N is only partially probed. Result 2’s decisive discriminator—the non-monotone $\hat{\alpha}_c(\tau)$ whose sparse-side peak $\tau^* \sim T/(Nr)$ *shifts with* N , so curves collapse on $\bar{n} = Nr\tau/T$ but split on $K = T/\tau$ —was not mapped here. The measured cells ($\tau = 125, 62.5$ at $N = 300, 600$) are all deep in the *dense* regime ($\bar{n} \sim 100\text{--}200 \gg 1$) and show only the rising/plateau segment ($2.76 \rightarrow 3.0$ as τ halves) with no size effect, which is consistent with the spectral axis but does *not* reach the sparse-side roll-off or test the N -coupling. The $\tau \rightarrow 0$ decline *rate* is likewise unverified (direction rigorous from the piecewise-constant structure, rate not).

O3. The absolute growth law is unresolved at this grid. The α -grid resolution is 0.4, and only two τ (hence two $K_{\text{eff}}^{\text{PR}}$ points, 6 and 12) were measured. This pins the *ordering* (smaller $\tau \Rightarrow$

larger $K_{\text{eff}}^{\text{PR}} \Rightarrow$ larger $\hat{\alpha}_c$) and the dt -flatness, but cannot separate the EVT $\ln \ln K$ form from a $\ln K$ or linear lift, nor resolve the absolute slope. As with the rest of the program, this is a *findable* capacity under a validated learner, not an existence statement.

Honest status. Status: PARTIAL—the dt -saturation (Result 1) is **SOLID**: pre-decided by a free \$0 witness ($\mathbb{E}[\max V]$ flat, $K_{\text{eff}}^{\text{rice}} \propto dt^{-0.513}$) and then confirmed by the learned $\hat{\alpha}_c$, which is identical (2.76) across a $2\times$ grid refinement while the Rice axis moves by $\sqrt{2}$. The boxcar is established as the program’s smoothness boundary, where the Rice axis is genuinely undefined and capacity reverts to the finite spectral/density axes (the #4 mechanism forced sharp). What remains *open*: the $\tau^*(N)$ sparse-side roll-off and its density-collapse (O2), the substitution of $K_{\text{eff}}^{\text{PR}}$ into the replica lift (O1), and the absolute growth law at finer α -resolution (O3).

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Theory derivation by a **statmech-theory** agent with SymPy spectral verification; the boxcar saturation is the sharp ($\tau_s = 0$) limit of the reduction #4 Gaussian-process saturation, whose OU cross-check was performed by codex (gpt-5.5, xhigh reasoning effort). Companion derivation and the SymPy-checked spectral identities are in `lit/tempotron-reductions/derivations/11-rect-capacity-prediction.md`; the kernel-spectrum reference is `lit/kernel-capacity-kb/C-kernels-and-spectra.md`; the Rubin–Monasson–Sompolinsky benchmark formulas, the rect $\leftrightarrow$ binned mapping, and the $K = T/\tau$ existence ladder are sourced and text-verified against the local PRL/arXiv in `lit/kernel-capacity-kb/D-sompolinsky-rect-benchmark.md`; the measured numbers are in `experiments/kernel_models_results.md` (per-kernel campaign plus the benchmark extension push).

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